



iCafe Cloud Management

OPERATOR & STAFF FIELD MANUAL

Official Reference for Secret Keys, Employee Management & Production Operations

SYSTEM ARCHITECTURE & PRODUCTION READINESS

- Cloud-Primary Engine: Supabase PostgreSQL database + 16 Deno Edge Functions + Vercel SPA.
- High-Availability Cafe Edge: Local LAN Express & SQLite server with 100% offline durability.
- Automated Verification: 659 passed regression tests (614 unit/regression + 45 cloud deployment).
- High-Security Kiosk: Win32 API low-level keyboard hook + Electron hardened sandboxed IPC.
- Financial Ledger: Transactional wallet deductions, atomic timer accounting & pause-aware billing.

VOLUME 1: KEYS & STAFF ONBOARDING

- Secret Emergency Station shortcuts (Quit, Lock, Unlock).
- Station Setup Master PIN security boundary.
- Admin Terminal privacy lock and operational hotkeys.
- Step-by-step Staff Employee invitation procedure.
- Front-desk Cashier, Manager & Admin permission roles.
- First-run mandatory credential change enforcement.

VOLUME 2: PRODUCTION & DAY-TO-DAY

- Step-by-step production deployment checklist.
- Handling day-to-day ISP Internet outages seamlessly.
- Sudden PC crash & power blink session recovery.
- Cashier shift handover & cash drawer reconciliation.
- The 4 Golden Rules: Deep Freeze, Static IP, Defender.
- Emergency station unpairing and troubleshooting.

DEPLOYMENT SPECIFICATIONS

Software Version: 1.0.0 (Release Candidate)

Local Database: SQLite 3 (WAL mode) / Cloud: Supabase PG 15

Customer Client: Electron 39.0.0 (NSIS Installer x64)

Default Master PIN: 062321

Default Member Password: 1234

Local Edge Fallback Port: TCP 3000

1. CUSTOMER STATION (CLIENT PC) SHORTCUTS

Kiosk & Emergency Controls

Key Combination	Action / Trigger	Description & Functional Purpose	Access Level
Alt + Shift + W	Emergency Station Quit	Prompts for Master PIN. Closes kiosk and cleanly exits Electron app.	High (Master PIN)
Alt + Shift + L	Emergency Station Lock	Prompts for Master PIN. Forces station immediately into locked kiosk mode.	High (Master PIN)
Alt + Shift + U	Emergency Station Unlock	Prompts for Master PIN. Unlocks station from locked state for staff testing.	High (Master PIN)
Right-Click (Timer)	Compact Timer Menu	Clicking the active compact HUD opens opacity settings (20%-100%) and tray options.	Customer Safe
Win Key / Win+D / Win+Tab	Windows Shell Chords	Swallowed by low-level Windows key hook (windows-key-hook.ps1). Prevents task switching.	Hardware Blocked
Alt+Tab / Alt+F4 / Ctrl+Esc	Application Escapes	Intercepted & cancelled by Electron before-input-event while station is locked.	System Blocked

2. ADMIN TERMINAL & MANAGEMENT SHORTCUTS

Cashier Console Controls

Key / Hotkey	Action / Trigger	Description & Functional Purpose	Category
Alt + Shift + L	Instant Privacy Lock	Locks Admin Terminal immediately. Requires Admin PIN or Password to unlock.	Console Security
Alt + Shift + U	Focus Unlock Field	Bypasses lock input trap to immediately focus the PIN/password box on lock screen.	Console Security
F8	Toggle UI Mode	Instantly switches between Simple Mode (Cashier POS) and Advance Mode (Owner).	Role Gated
F9	Toggle Theme Mode	Switches between standard Dashboard theme and Esports cyber gaming arena skin.	Theme Control
Escape	Dismiss Active Layer	Closes modals, slide-out drawer, mobile nav, or Quick Find popovers (disabled if locked).	Navigation
Enter (Quick Find)	Jump to Top Match	In header Quick Find bar, pressing Enter immediately routes to top matched PC or Member.	Productivity

CRITICAL CREDENTIALS & MASTER PINS REFERENCE

- Station Setup Master PIN (Default: 062321): Protects Server Connection settings, Emergency Quit/Lock/Unlock, and Station Launcher configuration. Verified using crypto.timingSafeEqual. Change via AEZAKMI_STATION_SETUP_MASTER_PIN.
- Default Member Password (1234): Assigned to newly created members. System enforces mandatory password update on login.
- Staff Invitation Temporary Password (Staff#XXXX): Random 4-digit code generated during staff onboarding.

STAFF MANAGEMENT & ROLE DELEGATION

Step-by-Step Operator Guide

The iCafe system features multi-tier staff role delegation, allowing cafe owners to grant cashier and supervisor permissions without compromising business revenue reports, rate plan tariffs, or master security credentials.

Role Name	Internal Key	Allowed Capabilities & Duties	Access Restrictions
Cashier / Front-Desk	cashier	Day-to-day front desk operations, member balance top-ups, walk-in guest passes, food/snack sales, shift reconciliation.	Tariffs, system settings, master PINs, employee invitations, financial audits.
Shift Manager	manager	Supervises cashiers, reviews daily shift handovers, handles station overrides, updates inventory stock quantities.	System rate pricing, business owner settings, developer console.
Branch Admin	admin	Full store authority: configures hourly rate plans, PC station layouts, invites new employees, manages store settings.	Multi-branch owner billing and developer platform configurations.

STEP-BY-STEP: ADDING AN EMPLOYEE TO THE SYSTEM

Admin Console Workflow

Step	Action	Operational Instructions & System Behavior
Step 1	Open Team Settings	Sign in as Owner or Admin. Press F8 to enter Advance Mode. Click Settings -> Team & Staff tab.
Step 2	Fill Employee Details	Enter Full Name (e.g. Sarah Jenkins), Email Address (e.g. sarah@icafe.ph), and Assigned Role.
Step 3	Click Invite Employee	System provisions an account in Supabase Auth / Local Edge with must_change_credentials = true.
Step 4	Credential Delivery	If Brevo API is set, an email with credentials is sent. Otherwise, copy the temporary password from the UI.
Step 5	First Login & Setup	Employee logs in using their email and temporary password (Staff#XXXX). Setup modal forces password change.
Step 6	Fast Terminal Unlock	Configure a 4-8 digit Cashier PIN in Settings -> Security & PIN for fast unlock without typing passwords.

FIRST LOGIN PASSWORD ENFORCEMENT (SECURITY FEATURE)

- When any employee logs in with a temporary password (Staff#XXXX), AdminCredentialSetup immediately locks the screen.
- The employee cannot dismiss the dialog with Escape or navigate away until they establish their own confidential password.
- In PIN mode, the cashier establishes a private numeric PIN for fast terminal access during busy gaming hours.

PRODUCTION ROLLOUT & ARCHITECTURE ROADMAP

Final Release Gates

Stage	Target Area	Description & Verification	Command / Action
1. Supabase Cloud Deploy	Database & Functions	Apply all 34 migrations and deploy 16 serverless Edge Functions to your Supabase project.	<code>npx supabase db push</code> <code>npx supabase functions deploy --use-api</code>
2. Cloud Admin Web Deploy	Vercel Deployment	Connect GitHub repo to Vercel. Set build command: <code>npm run build:admin</code> and configure public Supabase URL/key.	Vercel Dashboard / Git Push
3. Customer Station Installer	NSIS Packaging (x64)	Builds hardened Electron kiosk client into <code>apps/customer/installer</code> . Includes low-level key hook.	<code>npm --workspace apps/customer run dist:win</code>
4. Admin / Cafe Edge Installer	Native Rebuild & NSIS	Rebuilds native modules (better-sqlite3, argon2) for Electron ABI and packages desktop server bundle.	<code>npm --workspace apps/admin run dist:win</code>
5. Server PC LAN Inbound Rule	Windows Firewall	Allow TCP port 3000 inbound on Server PC so stations can fall back to local Cafe Edge during outages.	<code>New-NetFirewallRule -Port 3000 -Action Allow</code>
6. Station OTP Pairing	8-Digit One-Time Code	Generate code in Cloud Admin -> Clients -> Pair Station. Enter on customer PC to permanently bind.	Code format: XXXX-XXXX

PRODUCTION ENVIRONMENT VARIABLES REFERENCE

Configuration Guardrails

Variable Name	Scope / Target	Requirements & Safety Enforcement	Status
JWT_SECRET	<code>backend/.env</code>	Must be at least 32 characters in production. Backend throws fatal error if default.	Required
CORS_ORIGIN	<code>backend/.env</code>	Explicit comma-separated origins (e.g. <code>https://your-admin.vercel.app,http://localhost:5174</code>). No wildcard.	Required
AEZAKMI_STATION_SETUP_MASTER_PIN	<code>Server & Client env</code>	4-8 digits. Master recovery PIN. Default is 062321. Set custom PIN before deploying.	Recommended
BREVO_API_KEY	<code>Supabase & backend</code>	API key for transactional email delivery (quotations, invites, password resets).	Optional / Ready
AEZAKMI_CLOUD_ENABLED	<code>backend/.env</code>	Set to true on Cafe Edge server to enable automatic background replication to Supabase.	Required

AUTOMATED BACKUP & OBSERVABILITY GUARANTEE

- Database Backups: `backend/src/services/databaseBackup.js` takes automated verified SQLite backups every 6 hours.
- Retention: Keeps 28 point-in-time snapshots in `./data/backups` (7-day full rolling recovery).
- Observability: Local error logs in `./data/logs` with automatic sensitive parameter sanitization (passwords/PINs redacted).

DAILY OPERATIONAL FAILURE MODES & SAFETY NETS

Resilience Matrix

Operational Event	Automated System Safety Net	Cashier / Staff Action
ISP Internet Drop	Station detects WAN failure and routes requests to local LAN Cafe Edge (port 3000). Active sessions remain active.	No action needed. When WAN restores, syncWorker uploads backlog.
PC Crash / Power Loss	markSessionExit checkpoints remaining seconds into SQLite/Cloud. Station reboots into kiosk mode with time intact.	Customer logs back in to resume or cashier clicks Restore Interrupted.
Customer Lock / Break	Station lock freezes postpaid and prepaid billing timers. Paused duration is excluded from billing.	Customer unlocks with password; timer resumes from exact pause checkpoint.
Gamer Tamper / Cheat	Windows Key hook + Electron kiosk swallow Win+D, Alt+Tab, Alt+F4, Ctrl+Esc, and Task Manager.	Only staff with Master PIN (Alt+Shift+W) can exit or minimize the kiosk.
Cashier Shift Handover	Shift Management modal tallies Cash Top-ups, POS sales, guest passes, and flags drawer variance.	Cashier counts drawer cash, enters total, and closes shift report.

THE 4 GOLDEN RULES OF PHYSICAL CAFE DEPLOYMENT

Must-Follow Setup Rules

Rule	Detailed Instructions & Safeguards
Rule 1: Deep Freeze / Diskless Storage	Customer station stores its permanent paired credential in [InstallDir]/.aezakmi-customer right beside the exe. Install Customer Station on an unfrozen partition (e.g. D: drive or persistent game drive), or pair the station in Thawed mode before freezing the golden image.
Rule 2: Server PC Static IP	The server PC must have a static local IP address (e.g. 192.168.1.100) configured via Windows adapter settings or router DHCP reservation. This ensures client PCs always find the Cafe Edge fallback when internet drops.
Rule 3: Antivirus & Defender Whitelisting	Add the installation directories to Windows Defender exclusions on client PCs (C:\Program Files\iCafe Customer Station) and server (C:\ProgramData\iCafe Management System) so the keyboard hook is never blocked.
Rule 4: Master PIN Customization	Never leave the default Master PIN (062321) on public client stations. Set your own private 6-digit PIN in AEZAKMI_STATION_SETUP_MASTER_PIN on both server and clients.

EMERGENCY RECOVERY CHEAT SHEET

- Customer Station Stuck / Kiosk Escape: Press Alt + Shift + W and enter Master PIN (062321 or custom).
- Unpair / Re-pair Station: Click "Server" on Customer login screen -> Master PIN -> Server Connection -> Reset Pairing.
- Admin Console Locked: Press Alt + Shift + U to focus unlock box, then enter Admin PIN or Password.